# CURRICULUM VITAE

October 24, 2026


# David LACOSTE

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## EDUCATION

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| 2009 | Habilitation degree from P. et M. Curie University, Paris |
| 1996 – 1999 | Ph.D. in Theoretical Physics, Université Joseph Fourier, Grenoble |
| 1994 – 1995 | Master in Quantum Physics from Ecole Normale Supérieure, Paris |
| 1992 – 1995 | Undergraduate studies at Ecole Normale Supérieure, Paris |

## EMPLOYMENT

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| Jun 2016 – present | CNRS Directeur de Recherches (DR2 level) at Gulliver laboratory, ESPCI |
| Jun 2005 – Jun 2016 | CNRS Chargé de Recherches (CR1 level) at Gulliver laboratory, ESPCI |
| Jan 2002 – Jun 2005 | CNRS Chargé de Recherches (CR2 level) at Gulliver laboratory, ESPCI |
| Jan 2000 – Jan 2002 | Postdoctoral Fellow with T. Lubensky at Dept. of Physics, University of Pennsylvania in Philadelphia, working on various Soft Matter problems |

## HONORS

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| Sept 2020 | Distinction "Outstanding referee" for APS |
| Sept 2012 | Distinction "Outstanding referee" for EPL |
| 2011 – 2015 | Distinction "Prime d'excellence scientifique" from CNRS |

## RESEARCH

I am a theoretical physicist working on the statistical physics of living matter and on the emergence of life. My research spans stochastic thermodynamics, the origin of life, chemical reaction networks and autocatalysis, and the growth and fluctuations of cell populations. I am the author of 88 peer-reviewed publications (h-index 30), available on my web site `https://dlacoste22.github.io/publications.html` and on Google Scholar.

## SELECTED PUBLICATIONS


[1] A. Blokhuis, D. Lacoste and P. Nghe, *Universal motifs and the diversity of autocatalytic systems*, Proc. Natl. Acad. Sci. U.S.A. **117**, 25230–25236 (2020).

[2] L.-F. Rojas-Ochoa, D. Lacoste, R. Lenke, P. Schurtenberger and F. Scheffold, *Depolarization of backscattered linearly polarized light*, J. Opt. Soc. Am. A **21**, 1799–1804 (2004).

[3] F. Quemeneur, J. K. Sigurdsson, M. Renner, P. J. Atzberger, P. Bassereau and D. Lacoste, *Shape matters in protein mobility within membranes*, Proc. Natl. Acad. Sci. U.S.A. **111**, 5083–5087 (2014).

[4] A. W. C. Lau, D. Lacoste and K. Mallick, *Nonequilibrium fluctuations and mechanochemical couplings of a molecular motor*, Phys. Rev. Lett. **99**, 158102 (2007).

[5] M. F. Islam, K. H. Lin, D. Lacoste, T. C. Lubensky and A. G. Yodh, *Field-induced structures in miscible ferrofluid suspensions with and without latex spheres*, Phys. Rev. E **67**, 021402 (2003).


## ONGOING GRANTS

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| Sept 2026 – Aug 2029 | Coordinator of the MERCASTO project (PSL University), *Network-based detection of economic sub-structures responsible for carbon emissions*, in collaboration with S. Benoit (Université Paris Dauphine - PSL); funds the PhD of Jiwei Wang. |
| Oct 2025 – Sept 2029 | Coordinator of the ANR project *Porous life*, on the emergence of life from confined autocatalytic chemical reactions. |

## TEACHING IN THE LAST FIVE YEARS

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| **2025 – ongoing** | Tutorials on statistics, and lectures on energy transition for 1st year ESPCI students and master students |
| **2016 – ongoing** | Biophysics course for 3rd year students in ESPCI |
| **2023 – 2024** | Tutorials and lectures on Energy conversion in Paris-Saclay university |
| **2021 – 2023** | Organization and lectures for PSL weeks in ESPCI |
| **2021 – 2022** | Tutorials on Computation and Data Driven Biophysics at ENS |

## STUDENT SUPERVISION

I currently supervise 2 PhD students: B. Ledoux, who is in his last year, and Jiwei Wang, who started in September 2026. I have supervised 6 PhD students and about 9 postdocs so far.

## INVITED TALKS SINCE 2022

- Thermodynamics 2.0, Boone, NC, USA, August 2026
- IFOM workshop: From Cells to Populations: Quantitative Principles of Growth and Evolution, Milan, January 2026
- Conference on Extremal Statistics in Biology, Vienna, June 2025
- Workshop on Action vs. Inaction facing climate change, Zinal, Switzerland, May 2025
- Conference on biological information processing, Okinawa, Japan, April 2025
- Solvay Workshop in honor of P. Gaspard, Brussels, December 2024
- Evolution in Paris, Paris, September 2024
- Paris-Rice meeting, Paris, September 2024
- Thermodynamics 2.0 (online), Broomfield, USA, August 2024
- Workshop on Emergence of Information in Molecular Systems, MIAPP program, Garching, July 2024
- Workshop on Evolution of population dynamics, Pasteur Institute, Paris, June 2024
- Congress of the French Physical Society, Paris, July 2023
- Postmodern Thermodynamics conference, Luxembourg, December 2022
- Thermodynamics 2.0 (online), Riverside, USA, July 2022
- Stochastic Thermodynamics conference (online), ICTP, Bangalore, June 2022

## ACTIVITIES OF GENERAL INTEREST

- Member of two scientific committees of Paris Sciences et Lettres University (PSL), Qlife and Smart-waves, which concern biophysics and the physics of waves respectively.
- Member of committee "Objectif décarbonation" at ESPCI.
- Organization of conferences: 3 in 2017-2018; 2 in 2024, including a celebration of "Sadi Carnot's legacy" at Ecole Polytechnique, September 16-18, 2024; and 2 in 2026: a summer school on Pattern formation in biology in Beg Rohu, Brittany, in May 2026, and a conference on "Thermodynamics and Kinetics of Chemical Reaction Networks: from Cell Metabolism to Design Principles for Synthetic Life" in Garching in September 2026.